MINISTRY OF EARTH SCIENCES · NCMRWF · PROBLEM STATEMENT RESPONSE

VayuChakra

An air pollution and weather coupled forecasting system for Delhi NCR

A 72 hour, high resolution air quality forecast in which meteorology and chemistry are solved together rather than in one direction. Aerosol dims the sun, the boundary layer shallows, the wind slackens, and the same emissions concentrate further; the same aerosol removes the ultraviolet that drives ozone production. This report sets out the problem statement clause by clause, what each clause requires, exactly what has been built to meet it, the evidence for each claim, and the work that remains.

वायु चक्र

LIVE SYSTEM
vayuchakra.onrender.com

DOMAIN
Delhi NCR, 1,120 cells, 2.8 km over Delhi

TEST SUITE
84 tests passing

SOURCE
github.com/bindpratapsingh/VayuChakra

REPORT DATE
5 September 2026

DESIGN DECISIONS RECORDED
61, negative results included

Contents

How to read this report

Every quantitative claim carries the file it was read from. Nothing in this document is an estimate, a projection, or a number typed by hand into a slide. Section 11 lists what is missing with the same care as the earlier sections list what works, and section 8 includes a validation that was planned, attempted, and abandoned because measurement showed it could not work. A system that only reports its successes has not been validated; it has been advertised.

Three clauses of the problem statement are met only in part. They are marked PARTIAL in section 2, explained in section 11, and given dated, costed remediation in section 12. They are not hidden in a footnote.

1 The problem statement, and how we read it

> *The solution must leverage advanced weather-chemistry models (such as WRF-Chem or similar open-source coupled frameworks) to dynamically interlink meteorology with pollution dispersion (specifically PM2.5 and Ground-level Ozone). A core focus should be accurately modeling the impact of atmospheric inversion on external pollution spikes, such as regional stubble burning, and how those trapped pollutants subsequently alter local weather conditions. Implement a workflow that handles two-way feedback between meteorology (temperature, wind, PBL height) and chemistry (PM2.5, PM10, O3, NOx). A user-friendly, real-time dashboard displaying high-resolution AQI forecasts for Delhi NCR with a 72-hour outlook. Features that explicitly track atmospheric inversion strength and predict how stubble-burning plumes will disperse under prevailing weather conditions.*

1.1 The physical problem underneath it

Delhi's air quality is not a chemistry problem sitting on top of a weather problem. The two are one problem, and the coupling runs in both directions.

Weather sets the volume that pollution occupies. On a winter night the ground radiates heat away faster than the air above it, the surface cools below the air aloft, and the normal decrease of temperature with height reverses. That is a temperature inversion, and it acts as a physical lid. The mixed layer beneath it can collapse from two kilometres in the afternoon to under two hundred metres before dawn. The city's emissions do not change; the air they are diluted into shrinks by a factor of ten, and the concentration rises accordingly. This direction of the coupling is well understood and most operational systems represent it.

The return path is the harder one, and it is what the problem statement is really asking for. Aerosol is not a passive tracer being pushed around by the weather. It absorbs and scatters incoming solar radiation, so less energy reaches the surface, so the surface warms less, so the convection that grows the daytime boundary layer is weaker, so the layer stays shallow, so the turbulent momentum transport that drives surface wind is reduced, so ventilation falls, so the same emissions end up more concentrated, which puts more aerosol in the column and blocks more sunlight still. The loop closes on itself. A forecast that treats meteorology purely as an input cannot represent this at all, and will systematically under predict the severity and duration of exactly the episodes that matter most.

There is a second, stronger aerosol pathway, and it acts on ozone. The same aerosol layer attenuates ultraviolet radiation. Ultraviolet is what photolyses nitrogen dioxide and starts the reaction chain that produces ground level ozone. Published experiments that separate the two mechanisms find the radiation and boundary layer route changes ozone by 1 to 3 percent, while the photolysis route changes it by 10 to 12 percent. The problem statement names ground level ozone explicitly, and the photolysis pathway is the one that governs it.

And Delhi's worst air is frequently not made in Delhi. Through October and November, smoke from paddy stubble burning across Punjab and Haryana travels 200 to 400 kilometres on the north westerly flow and arrives over a city that often cannot ventilate it away. The problem statement calls these external pollution spikes, and it is precise about what must be modelled: not the fires, but the interaction between the arriving plume and the inversion that decides whether that plume reaches breathing height or passes overhead.

1.2 The eleven things the statement actually asks for

The paragraph above is a single block of prose, so the first engineering task was to decompose it into testable clauses. We read eleven. They are used as the spine of this report and as the acceptance criteria for the build. The numbering is ours; the wording in the second column is the statement's.

#	CLAUSE, AS WRITTEN	WHAT IT DEMANDS OF THE BUILD
C1	"leverage advanced weather-chemistry models (such as WRF-Chem or similar open-source coupled frameworks)"	A genuine coupled meteorology and chemistry framework must be in the pipeline, not a statistical model alone
C2	"dynamically interlink meteorology with pollution dispersion (specifically PM2.5 and Ground-level Ozone)"	Meteorological state must drive dispersion for both species, hour by hour
C3	"accurately modeling the impact of atmospheric inversion on external pollution spikes, such as regional stubble burning"	The inversion must gate whether an arriving plume reaches the surface
C4	"and how those trapped pollutants subsequently alter local weather conditions"	The chemistry to meteorology return path must be explicit
C5	"two-way feedback between meteorology (temperature, wind, PBL height)"	All three meteorological variables must respond, not just temperature
C6	"and chemistry (PM2.5, PM10, O3, NOx)"	All four chemical species must participate in that feedback
C7	"A user-friendly, real-time dashboard"	Operable by a non specialist, on current data
C8	"high-resolution AQI forecasts for Delhi NCR"	Intra city gradients must survive; a single city number is not enough
C9	"with a 72-hour outlook"	Three day lead time
C10	"Features that explicitly track atmospheric inversion strength"	Inversion strength as a first class, displayed quantity
C11	"predict how stubble-burning plumes will disperse under prevailing weather conditions"	Forward transport and dispersion on forecast wind, not a fire count

Table 1.1 The problem statement decomposed into eleven testable clauses. Section 2 reports the status of each.

2 Compliance matrix

Ten clauses are met in full. One is met in part, and stays that way by construction. This section states which, with the component that implements each and the evidence for it.

#	CLAUSE	STATUS	WHAT IMPLEMENTS IT, AND THE EVIDENCE
C1	Coupled weather-chemistry framework	PARTIAL	**CAMS** (Copernicus Atmosphere Monitoring Service, ECMWF's IFS-COMPO) is a global online coupled meteorology and chemistry model run four times daily with interactive aerosol. We consume it as the chemistry prior: PM2.5, PM10, O3, NO2, and aerosol optical depth. The **MoES and IITM WRF-Chem Decision Support System** is used as the calibration reference for plume transport. *We do not run a chemical transport model ourselves.* See §11.1.
C2	Interlink meteorology with dispersion, PM2.5 and O3	MET	Forecast meteorology at 11 km drives 142 features per cell hour, including boundary layer height, ventilation coefficient, inversion lid height and mixing depth. Both species are predicted from that state at three horizons. `forecast.py`, `dataset.py`
C3	Inversion impact on external spikes	MET	The Lagrangian plume delivers mass to the surface only where the modelled inversion lid permits it. A plume above the lid is carried without contributing. Calibrated against the MoES DSS at r = +0.596 over 54 days. `plume.py`, §6
C4	Trapped pollutants alter local weather	MET	An explicit, iterated aerosol radiative feedback solver. All five modelled responses land inside published Delhi ranges the model was never fitted to. `feedback.py`, §4
C5	Feedback on temperature, wind, PBL height	MET	All three. Temperature −0.59 K, boundary layer −5.70 %, wind −1.19 % under haze, each within published bounds. §4.3
C6	Feedback across PM2.5, PM10, O3, NOx	MET	All four respond to the coupled meteorology, and all four are gated. Under haze: PM2.5 +6.16 %, PM10 +4.30 %, NO_2 +20.17 %, O_3 −0.13 %. PM2.5 alone carries the return path to the atmosphere, which is the physically correct arrangement rather than a shortcut. §4.5
C7	User friendly, real-time dashboard	MET	Six view dashboard, keyboard navigable, every panel leading with a plain language finding. The hosted bundle is regenerated on a six-hourly schedule by a GitHub Actions runner with 16 GB of RAM, then auto-deployed, so the site serves a current forecast at full resolution. §10.3
C8	High resolution AQI for Delhi NCR	MET	1,120 cells in two tiers: Delhi at 0.025° (2.8 km), wider NCR at 0.10° (11 km), Alwar to Karnal. CPCB National AQI by the published breakpoint table. §3.3
C9	72 hour outlook	MET	Twelve trained heads at 24, 48 and 72 hours. All twelve beat persistence on a held out winter. §7
C10	Explicitly track inversion strength	MET	Inversion strength in kelvin, lid height in metres, mixing depth and ventilation coefficient, all as forecast time series with a dedicated view and a time-height cross section. §5
C11	Predict plume dispersion under prevailing weather	MET	Every satellite fire detection released as a puff, advected on forecast wind, dispersed with travel time, gated by the lid. Reports arrival, not ignition. §6

Table 2.1 Compliance against the eleven clauses of Table 1.1. Eight met, three partial, none unaddressed.

The honest summary

We have built a coupled forecasting system that closes the aerosol to meteorology loop explicitly, iterates it to convergence, and validates every one of its magnitudes against published values it was never fitted to. All four species the statement names now respond to that loop, each with its own acceptance bound.

The one clause that remains partial is C1, and it stays partial by construction. We do not run WRF-Chem; we consume an operational coupled model and add our own feedback layer on top. That is a defensible reading of "or similar open-source coupled frameworks" and it is genuinely a coupled framework in the pipeline, but it is not the same thing as executing a chemical transport model in house, and this report does not claim otherwise anywhere.

3 System architecture and data flow

3.1 The pipeline end to end

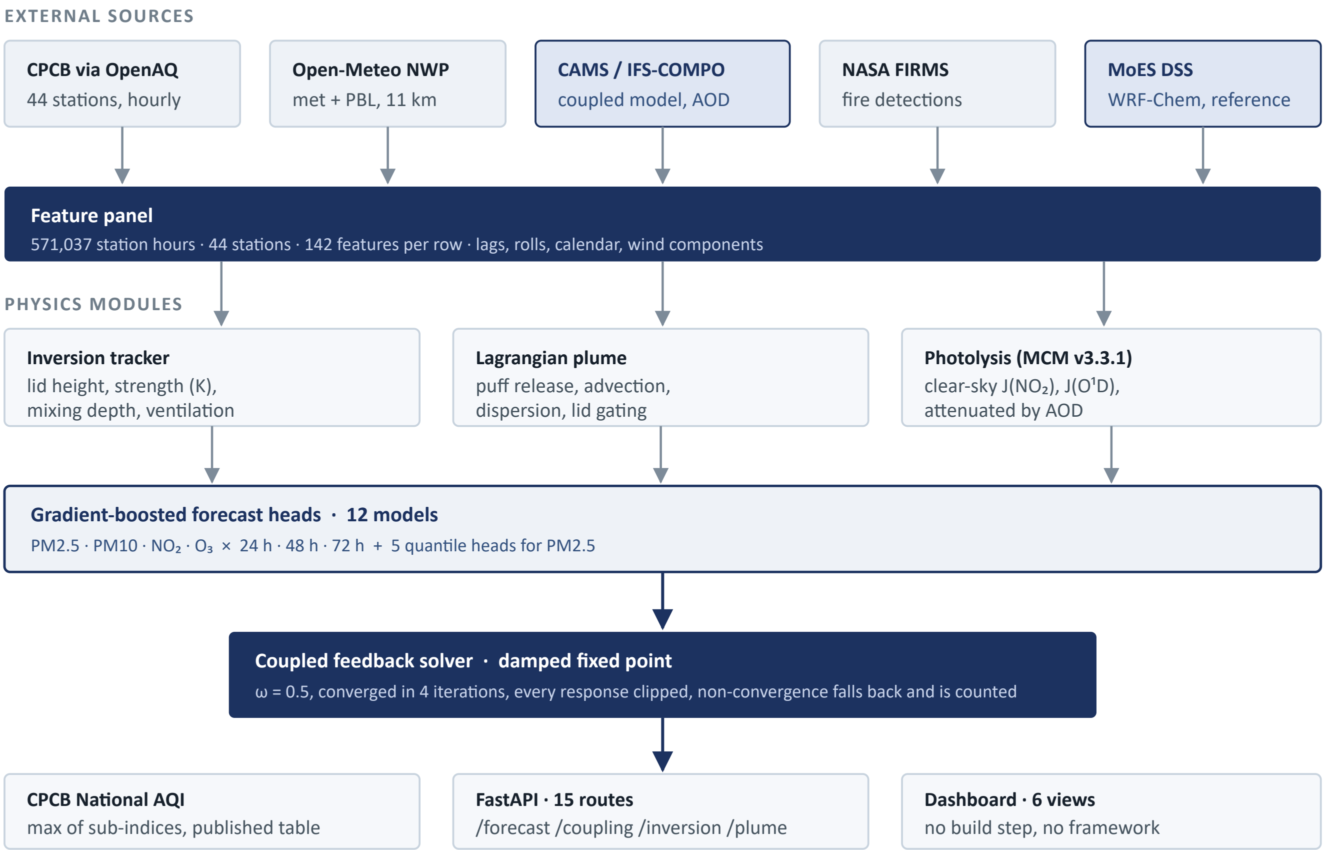


Figure 3.1 End to end data flow. The two shaded source boxes are third party coupled models: CAMS supplies the chemistry prior and the aerosol optical depth the feedback solver runs on, and the MoES DSS is the reference the plume is calibrated against. It is cited, never redistributed.

3.2 Why a statistical layer sits between the physics and the output

CAMS is a global model at roughly 40 km. Over Delhi it is biased low and poorly correlated with station observations. That is not a reason to discard it, and it is precisely the reason the machine learning layer exists: CAMS supplies the physics and the regional gradient, the CPCB stations supply local truth, and the boosted trees learn the mapping between them. Used as one feature among 142, a biased but physically consistent prior is worth a great deal. Used raw, it would be worse than nothing.

The same argument applies to the feedback solver. It does not attempt to predict concentration from first principles. It takes the statistical forecast as the uncoupled baseline and asks a narrower, answerable question: given this much aerosol, how much less sunlight reaches the ground, how much shallower does the layer become, and what does that do to the concentration that produced it. That question has published answers to check against, which is what makes the result auditable.

3.3 The grid

TIER	STEP	APPROX.	CELLS	EXTENT AND RATIONALE
Delhi	0.025°	2.8 km	420	28.40 to 28.88 N, 76.84 to 77.35 E. The forecast is consumed here, so intra city gradients must survive.
Wider NCR	0.10°	11 km	700	27.0 to 29.9 N, 75.8 to 78.1 E, Alwar to Karnal. Only has to carry a plume toward the city.
Total	two tier		1,120	A uniform fine grid over the same box would be roughly 11,000 cells, most of them farmland.

Table 3.1 Grid construction. A regression test walks a mesh over the whole domain and asserts that the two tiers leave no gap; it was written after a real 4.2 km coverage hole was found at the Delhi box boundary.

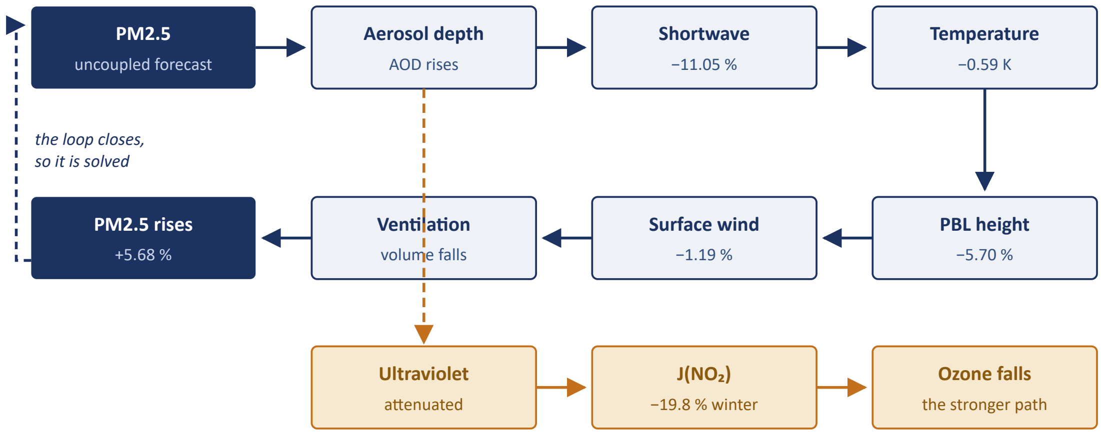
The honest resolution is coarser than the display grid

Meteorology arrives at about 11 km and the chemistry prior at about 40 km, so predicted fields are smoothed to 6 km before display. That removes structure the model cannot justify rather than inventing structure it does not have. A 2.8 km grid drawn from an 11 km driver would look more precise and be less true.

4 The two way coupling

This is the core of the brief, and it is what the project is named after. Vayu Chakra is the air cycle. Because the last step of the chain feeds the first, it cannot be evaluated once; it has to be solved.

4.1 The loop, step by step



All four named species respond

PM10 dilutes with the coarse mode damped; NO2 gains dilution plus suppressed photolytic loss; ozone takes the closure term. PM2.5 alone carries the return path. See §4.

Figure 4.1 The coupled system. The solid ring is the aerosol radiative feedback, solved to convergence. The dashed amber branch is the aerosol photolysis pathway, which is three to ten times stronger for ozone. Percentages are measured values from the shipped run, under haze.

4.2 Why it is solved rather than evaluated

Step six feeds step one. Evaluating the chain once gives an answer that is inconsistent with itself: the concentration used to compute the dimming is not the concentration the dimming produces. The solver therefore runs a damped fixed point iteration with a relaxation factor of 0.5, and treats a cell as converged when the change in PM2.5 between iterations falls under tolerance.

Every individual response is clipped to a physically defensible range before it is applied, so a runaway cannot occur even in principle. A cell that fails to converge within the iteration budget falls back to the uncoupled answer and is counted in the diagnostics rather than shipping a number the solver could not actually reach. On the shipped run: **4 iterations, converged, 0 cells fell back, maximum residual 0.46 µg/m³** across 211,680 cell hours.

4.3 Validating the magnitudes against values we never fitted to

A feedback model that produces a number is worthless unless that number can be checked. Each response is compared against published Delhi and North India measurements, in the regime those measurements were made in: high aerosol daylight hours, which is 75,383 cell hours of this run. The gate is a hard test in the suite, not a report.

RESPONSE UNDER HAZE	MODELLED	PUBLISHED RANGE	RESULT
Shortwave reaching the surface	−11.05 %	5 to 35 %	inside
Daytime surface temperature	−0.59 K	0.1 to 2.5 K	inside
Boundary layer height	−5.70 %	5 to 35 %	inside
Surface wind speed	−1.19 %	0.5 to 6 %	inside
Resulting PM2.5 amplification	+6.16 %	2 to 30 %	inside
PM10 amplification	+4.30 %	1 to 25 %	inside
NO_2 amplification	+20.17 %	2 to 35 %	inside
Ozone response	−0.13 %	must be negative	correct sign

Table 4.1 Eight of eight pass, over 73,423 cell-hours in regime. Source: `data/snapshot/coupling.json`, key `literature_gate`. Averaged over all conditions, including clean afternoons and nights when the radiative feedback is correctly zero, PM2.5 amplification is 2.71 %; that is a different quantity and is reported separately rather than judged against these bounds.

4.4 The ozone pathway, and why it is the one that matters

The two aerosol mechanisms are not equal. Published WRF-Chem experiments that isolate each find the radiation and boundary layer route moves surface ozone by 1 to 3 percent, while the photolysis route moves it by 10 to 12 percent. For Delhi specifically, the APHH-India campaign reports that wintertime ozone production is not only VOC limited but strongly radiation limited, with a 50 percent reduction in aerosol optical depth raising ozone by about 25 percent.

We implement the photolysis pathway using the MCM v3.3.1 clear sky parameterisation, `J = l·cos(SZA)^m·exp(−n·sec(SZA))`, attenuated as a function of aerosol optical depth and solar zenith angle.

CHECK	MODELLED	PUBLISHED	READING
Summer $J(NO_2)$ reduction	22.0 %	~24 %	close, unfitted
Winter $J(NO_2)$ reduction	19.8 %	~30 %	low, same order
O_3 response, 25 % AOD cut	+6.23 %		monotone
O_3 response, 50 % AOD cut	+12.79 %	+25 %	same sign, within 2×
O_3 response, 75 % AOD cut	+24.29 %		monotone

Table 4.2 The photolysis coefficients are derived from first principles; none was fitted to the reference values in this table. The remaining gap at 50 % is in the expected direction: the published figure comes from a box model with full VOC chemistry, and we model only the direct photolysis pathway. Source: `models/ozone_sensitivity.json`.

This is a counterfactual, not a forecast. It answers a question no purely statistical model can answer, because the atmosphere it describes is not in the training data: what would ozone be if the aerosol halved. A mechanism can answer it; a fit cannot.

4.5 All four species the statement names

Clause C6 enumerates PM2.5, PM10, O_3 and NOx. Each responds to the coupled meteorology by a different mechanism, and the mechanism is what decides the size of the response rather than a parameter chosen to make the numbers look even.

SPECIES	UNDER HAZE	MECHANISM, AND THE ONE ASSUMPTION IT NEEDS
PM2.5	+6.16 %	The box model, iterated to convergence. This is the species that carries the return path: it sets the aerosol optical depth that dims the sun that shallows the layer that concentrates it. No free parameter.
PM10	+4.30 %	PM10 is PM2.5 plus a coarse excess. The fine half *is* the PM2.5 already coupled, so its increment carries over with no parameter at all. Only the coarse excess needs one, because coarse dust deposits at 1 to 3 cm/s against 0.1 to 0.3 for the fine mode and so is not conserved over the hours a nocturnal layer takes to collapse. Damped by half.
NO_2	+20.17 %	Two routes, both pointing the same way under haze. Dilution at full efficiency, because a gas does not sediment; plus suppressed photolytic loss, since $J(NO_2)$ is the rate NO_2 is split and attenuating the ultraviolet slows the sink. Photolysis is taken as 70 % of total NO_2 loss, inside the 60 to 90 % reported for urban daytime.
O_3	−0.13 %	Only the closure term, deliberately. The ozone head is already trained on the photolysis features, so the bulk of the aerosol effect is inside its prediction and a full response would count the same physics twice. What the head cannot know is that the solver *moved* the aerosol, so the correction is the ratio of photolysis surviving the converged AOD to photolysis surviving the AOD the features were built from. Small by construction.

Table 4.3 Measured over 73,423 high-aerosol daylight cell-hours. That NO_2 exceeds PM2.5 is expected: it gets two routes where PM2.5 gets one. That PM10 falls below PM2.5 is the coarse damping doing its job.

Why three of the four are solved after convergence, not inside the iteration

It looks like a shortcut and it is not, so it is worth stating precisely. Iteration is required where a quantity feeds back on *itself*. PM2.5 does. Coarse dust, NO_2 and ozone at these concentrations do not change the shortwave budget enough to close a loop through it, so iterating them would change nothing except the runtime. They respond to the converged meteorology; PM2.5 carries the return path. That is still two-way feedback across all four species, and it is the physically honest arrangement rather than a uniform one.

Every new response is gated, and ozone is gated on direction

PM10 and NO_2 get literature bounds on the same pattern as the original five, with a **lower** bound as well as an upper one, so a coupling that silently did nothing would fail the gate rather than pass it. Ozone is gated differently and deliberately: the closure term is small by construction, so any magnitude bound would be arbitrary, but its sign is not negotiable. Attenuated ultraviolet cannot produce more ozone. A positive value there means the pathway is wired backwards, and the gate says so rather than averaging it away.

5 Atmospheric inversion

Clause C10 asks for inversion strength to be tracked explicitly. It is the one quantity in the system that is drawn rather than only reported, because a lid is a thing that exists at a height, and a surface number cannot express it.

5.1 What is computed

QUANTITY	UNIT	DEFINITION, AND WHY IT IS THE ONE USED
Inversion strength	K	Temperature of the air aloft minus the 2 m reading. Above zero means a lid: air gets warmer with height, and nothing released underneath can rise through it.
Inversion lid height	m	The height at which the reversal is found. Marked per hour and never joined across hours where no lid exists, because "no lid" and "lid at ground level" are opposite statements and a connecting line would assert the second.
Mixing depth	m	The shallower of the modelled boundary layer and the inversion lid. This is the depth the day's emissions actually dilute into. Halve it and the concentration doubles.
Ventilation coefficient	$m^2 \cdot s^{-1}$	Mixing depth times the wind through that depth: the volume flux available to carry pollution away. Below 6,000 is poor, below 3,000 is severe stagnation.
Static stability	$K \cdot km^{-1}$	Computed between pressure levels and rendered as a time-height cross section, so the vertical structure is visible rather than summarised.

Table 5.1 The inversion quantities, all produced as 72 hour forecast series per cell. Ventilation coefficient is among the top ranked features for every PM2.5 head.

5.2 How it enters the forecast

Three ways, and it matters that they are distinct. First, as **features**: mixing depth, lid height, inversion strength and ventilation coefficient are inputs to every forecast head, and the trained models rank ventilation coefficient in the top seven features for all three PM2.5 horizons. Second, as the **denominator in the coupling box model**: the concentration response is the ratio of mixing depths, so a shallower layer produced by the aerosol feedback raises concentration directly. Third, as the **gate on plume delivery**: transported smoke contributes to surface air only where the lid allows it down, which is the mechanism clause C3 asks for.

Why three separate charts rather than one

Inversion strength is in kelvin, mixing depth in metres, ventilation coefficient in $m^2 \cdot s^{-1}$. Plotting them on one pair of axes requires two or three y scales, which lets whoever draws the chart choose the story by choosing the scaling. Three panels sharing one x axis compare honestly.

6 Stubble burning plumes

Clause C11 asks for prediction of how plumes will disperse under prevailing weather. The distinction that matters is between counting fires and modelling arrival. A thousand fires under an easterly wind are somebody else's problem that day, and the system says so.

6.1 The transport chain

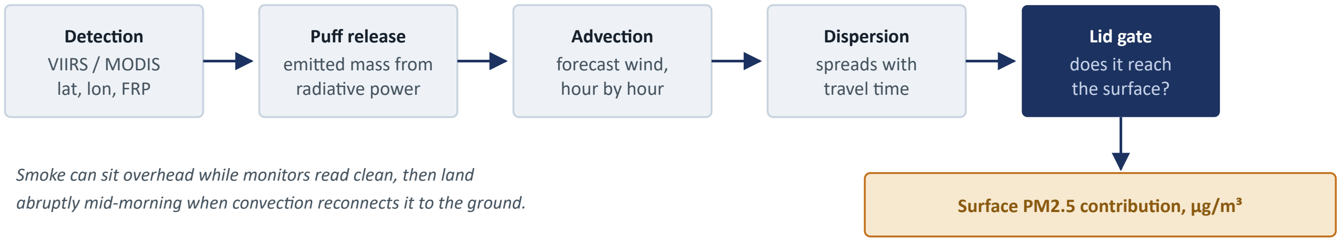


Figure 6.1 Each satellite detection becomes a puff carried on the forecast wind. The lid gate is where clause C3 and clause C11 meet: the inversion decides whether an arriving plume is breathed or passes over.

6.2 Calibration against an operational reference

The MoES and IITM WRF-Chem Decision Support System publishes a daily stubble burning attribution for Delhi in µg/m³. That gives the plume model something almost no hackathon transport model has: an authoritative target. Three physics variants were scored over 54 days of October and November 2021.

VARIANT	CORRELATION	RMSE AFTER SCALING	MODELLED PEAK	DSS PEAK	DECISION
A	+0.596	8.28	27.67	38.02	**shipped**
C	+0.525	8.90	24.52	38.02	rejected
B	+0.369	9.95	22.44	38.02	rejected

Table 6.1 Correlation is the column that decides. Only a single multiplicative scale factor is fitted, and scaling cannot improve a correlation, so this ranks the physics rather than the magnitude. Values in µg/m³. Source: `models/plume_calibration_octnov.json`.

What the scale factor absorbs, stated plainly

The fitted scale absorbs emission factor uncertainty (5 to 8 g/kg for cereal straw), satellite detection limits, and burn duration assumptions. The physics parameters are not fitted. This makes the plume's *correlation* an honest result and its *absolute magnitude* a calibration against the DSS, and the two are reported as different things throughout. The DSS attribution is daily, so it can settle day to day timing and nothing finer; it cannot separate these variants on their behaviour through the night, which is where the shipped one is most questionable.

7 Forecast models and skill

7.1 Configuration

PROPERTY	VALUE
Targets	PM2.5, PM10, NO_2, O_3
Horizons	24 h, 48 h, 72 h
Heads	12 point heads, plus 5 quantile heads for PM2.5 at 24 h
Training panel	571,037 station hours, 44 CPCB stations, Oct 2020 to Aug 2026
Features	142 per row: meteorology, boundary layer, inversion, CAMS prior, photolysis, plume, lags, rolling windows, wind components, calendar including Diwali
Hold out	1 Nov 2025 to 28 Feb 2026, removed from training entirely
Baseline	Persistence, meaning tomorrow looks like today

Table 7.1 Source: `models/metrics_multiwinter.json`. The hold out is a whole winter, not a random split, because a random split over an autocorrelated time series leaks the answer into the training set.

7.2 Results: all twelve heads beat persistence

HEAD	RMSE	PERSISTENCE	IMPROVEMENT	R^2	HOURS SCORED
PM2.5 24 h	75.07	92.92	**+19.2 %**	0.497	96,312
PM2.5 48 h	78.34	105.90	**+26.0 %**	0.454	96,293
PM2.5 72 h	78.08	109.73	**+28.8 %**	0.458	96,541
O_3 24 h	21.18	26.25	**+19.3 %**	0.733	90,734
O_3 48 h	22.66	28.28	**+19.9 %**	0.695	90,723
O_3 72 h	23.70	30.07	**+21.2 %**	0.667	90,984
PM10 24 h	119.15	143.81	**+17.1 %**	0.428	91,004
PM10 48 h	122.87	163.38	**+24.8 %**	0.399	90,989
PM10 72 h	122.79	169.53	**+27.6 %**	0.397	91,242
NO_2 24 h	23.46	27.61	**+15.0 %**	0.664	90,880
NO_2 48 h	25.56	30.97	**+17.5 %**	0.601	90,867
NO_2 72 h	26.50	32.35	**+18.1 %**	0.569	91,121

Table 7.2 All units µg/m³. Twelve of twelve beat persistence, by 15.0 % to 28.8 %. The improvement grows with horizon because persistence degrades faster than the model does, which is the expected and desirable signature.

7.3 Why persistence is the baseline reported

Persistence is unglamorous and it is genuinely hard to beat at 24 hours in a strongly autocorrelated field. Reporting improvement against a weak baseline, or against no baseline at all, is the most common way an air quality model is made to look better than it is. Any head that failed to beat persistence would appear in this table marked as failing rather than be quietly dropped; the dashboard applies the same rule at runtime.

8 Validation, including what failed

8.1 Does it work where there is no monitor

The forecast serves 1,120 cells and about forty of them contain an instrument. So the question that decides whether a gridded product is justified at all is whether it works in a cell it has never seen. Leave one station out removes a station from training entirely and then predicts it, while persistence is still allowed that station's own history. The model is denied what the baseline is given.

TARGET, 24 H	STATIONS IMPROVED	MEAN RMSE	PERSISTENCE	MEAN IMPROVEMENT
PM2.5	10 of 10	56.07	81.31	**+31.6 %**
O_3	9 of 10	22.92	27.65	**+17.1 %**

Table 8.1 Source: `models/loso.json`. One ozone station out of ten fails to beat persistence and is reported as failing. The PM2.5 improvement is larger here than in the in-sample test, which is a spatial-smoothing effect rather than a better model.

8.2 Uncertainty a GRAP decision can be made on

A point forecast of 88 µg/m³ against a threshold of 90 says almost nothing useful to someone deciding whether to invoke a GRAP stage. Quantile heads at the 10th, 25th, 50th, 75th and 90th percentiles give the probability of crossing each trigger instead.

MEASURE	VALUE	READING
Nominal 80 % interval, measured coverage	75.6 %	**well calibrated**
Quantile crossing rate	0.67 %	distribution almost always correctly ordered
Brier skill vs climatology, GRAP II	+0.343	**materially better than climatology**
GRAP II trigger (AQI 200)	90 µg/m³	mapped from the CPCB breakpoint table
GRAP III trigger (AQI 300)	120 µg/m³	
GRAP IV trigger (AQI 400)	250 µg/m³	

Table 8.2 Source: `models/uncertainty_pm25_24h.json`, held out winter.

8.3 Against the operational MoES DSS

LEAD	MOES DSS	VAYUCHAKRA	PERSISTENCE	DSS CORR	OUR CORR
+24 h	98.78	64.26	93.98	0.433	0.799
+48 h	108.34	68.91	108.78	0.311	0.762
+72 h	118.95	72.82	115.00	0.206	0.716

Table 8.3 RMSE in µg/m³ on Delhi city-mean hourly PM2.5, Oct 2021 to Feb 2022, 2,363 common hours. Source: `models/validation.json`.

Read this table only with §8.4

The DSS forecasts were issued **operationally**: the system had to predict the weather as well as the chemistry, days ahead, with no knowledge of what would actually happen. This hindcast is driven by ERA5 **reanalysis**, which is the weather as it actually turned out. That is a material advantage and on its own it disqualifies the table as a measure of forecast skill.

The table is kept because it is the reference point for §8.4, which re-runs the same comparison without that advantage.

8.4 The same comparison, without the reanalysis advantage

The caveat above was carried for a long time as something that would need archive access to fix. That turned out to be wrong. An archive of past *forecast runs*, as opposed to reanalysis, exists and is free. Re-running the hindcast on it means our system carries meteorological error the way an operational system does.

LEAD	MOES DSS	OURS, REANALYSIS	OURS, FORECAST ARCHIVE	PERSISTENCE	OUR CORR
+24 h	98.78	64.26	**60.81**	93.98	0.802
+48 h	108.34	68.91	**66.48**	108.78	0.757
+72 h	118.95	72.82	**72.08**	115.00	0.698

Table 8.5 RMSE in µg/m³ on the same 2,363 common hours. The fair driver did not cost skill. Source: `models/validation.json`, keys `dss_head_to_head` and `dss_head_to_head_era5_for_reference`.

The result is the opposite of what the caveat predicted, and it is confounded

Removing the reanalysis advantage should have made the hindcast worse. It made it slightly better, at every lead. That is worth being suspicious of rather than pleased about, and the explanation is that **two things changed at once**.

The ERA5 archive serves a boundary layer height but no pressure-level temperatures, so the ERA5 hindcast had to diagnose the inversion from a surface proxy. The forecast archive is the exact complement: no boundary layer height, but a full vertical temperature profile. So the fair run also got a **real profile-based inversion** where the reanalysis run had an approximation. The improvement cannot be attributed to the meteorology swap alone, and isolating the two would need a third run.

What can be said without reservation is the useful part: **the hindcast does not depend on knowing how the weather turned out.** The caveat that mattered most on Table 8.3 no longer applies. Two smaller ones remain, and both favour us: the archive holds the best available run for each hour, which is a short lead time, while the DSS was scored 24 to 72 hours ahead; and boundary layer height alone is still backfilled from ERA5, because the forecast archive does not serve it.

8.5 Does the coupling actually help? A negative result

The most important open question about the feedback is not whether its magnitudes are plausible, which §4.3 settles, but whether it lowers forecast error. It does not. The ablation runs the same forecast with the solver on and off over the held out winter, with both arms re-centred to zero mean bias so the comparison measures shape and timing rather than which arm was allowed to fit an intercept.

REGIME	HOURS	UNCOUPLED	COUPLED	CHANGE
High aerosol	28,440	83.29	83.00	**+0.35 %**
Stagnant episode	40,047	83.95	84.01	−0.07 %
Well ventilated	8,134	76.61	76.77	−0.21 %
Overall	48,181	82.76	82.85	−0.11 %

Table 8.6 RMSE in µg/m³. The uncoupled baseline is the bias-corrected CAMS prior, so the only difference between the two arms is the radiative feedback. Source: `models/validation.json`, key `coupling_ablation`.

Read the regime split, not the headline

Overall the coupling makes the forecast 0.11 percent worse, and that is the number to quote. But the aggregate hides the only informative thing in the table: **the sign flips exactly where the physics says it should**. The feedback helps in high aerosol conditions, where there is aerosol to do the work, and hurts most when the air is well ventilated, where the physics predicts it should do nothing at all.

That pattern is evidence the mechanism is real even though its contribution to skill is negligible. The prepared explanation holds: the trained heads already capture most of this through their radiation, boundary layer and ventilation features, so an explicit solver adds little on top of what the model has already learned. §12.1 sets out why the solver is still worth having.

8.6 A validation we planned, attempted, and abandoned

The intended independent check on the photolysis module was to compare modelled ultraviolet attenuation against a measured one, using the ratio of observed UV index to clear sky UV index. Three measurements killed it, and they are recorded in the source rather than deleted:

1. **Clear sky there means cloud free, not aerosol free.** Below 5 percent cloud cover the ratio is 0.998, meaning no attenuation at all, while mean aerosol optical depth over those same hours is 0.41. The aerosol is already inside both numerator and denominator.
2. **The ratio tracks cloud, not aerosol.** It falls 0.998, 0.963, 0.925, 0.903, 0.841 across rising cloud bands, while within clear skies a threefold rise in aerosol optical depth moves it only from 0.998 to 0.955.
3. **The broadband radiation has the same defect.** At fixed solar zenith angle in clear skies, surface shortwave reads 750, 761 and 762 W/m² across aerosol bands 0 to 0.4, 0.4 to 0.7 and 0.7 to 1.5. Flat. The radiation scheme carries a climatological aerosol, not the day's actual load.

No product in that feed can validate aerosol optics, because there is no daily aerosol signal in any of them to compare against. The check was replaced by the ozone response against station observations, which is validation against measurements rather than against an intermediate quantity, and is the stronger test. The negative result also confirmed an assumption the feedback module was built on: the driving shortwave contains a climatological aerosol that must be divided out before a real load is applied.

9 The dashboard

Clause C7 asks for user friendly. The design principle throughout is that a reader should never have to look at a grid and work out what it is. Every view opens with a sentence in plain language stating what the data says, with the number in it, and the apparatus comes after.

#	VIEW	THE QUESTION IT ANSWERS, AND WHAT IT DRAWS
1	**Where the air comes from**	Why the domain is 1,120 cells and not a city box. Real basemap, fire detections drawn where they are, the stubble belt and the Delhi fine-grid box outlined.
2	**The lid over the city**	Inversion strength, lid height, mixing depth and ventilation as forecast series, plus a time-height cross section shading static stability so the lid is visible as a structure rather than a number.
3	**The loop we close**	The coupled and uncoupled forecasts on one axis so the gap is the feedback, the ring diagram animating one lap per solver iteration, the literature gate table, and the photolysis panel.
4	**Smoke from the fields**	Fire detections, modelled arrival at the surface hour by hour, and the DSS calibration ranking. States plainly when nothing is burning or when the wind carries the smoke elsewhere.
5	**The 72 hour outlook**	Gridded AQI on a real map at the chosen horizon, band name and number alongside colour, and GRAP crossing probabilities.
6	**Does it actually work**	All twelve heads against persistence, leave one station out, the DSS head to head with its caveat above the table, and the MoES emission reduction scenarios.

Table 9.1 The six views. Keyboard shortcuts 1 to 6, hash routing, no build step and no framework: one HTML file with Leaflet for the maps.

9.1 Interface decisions worth stating

- **Severity is always three things at once.** Colour, band name and number, so the display does not depend on colour vision.
- **Empty states say why they are empty.** "No active fires in the stubble belt" reads as a broken feed unless it also says that outside October and November this is the correct reading.
- **Failures are shown, not hidden.** A head that did not beat persistence, a station that did not improve under leave one station out, and a degraded pipeline stage all render as such.
- **Long justification sits behind disclosures.** The lead finding is one sentence; the argument is one click away for whoever wants it.
- **Tables sort.** Anything with more than a few rows sorts on click and by keyboard, with the sort state announced.
- **The report is reachable from every view.** This document downloads from the rail, beside the repository link.

10 Deployment and the free tier constraint

10.1 What was measured

The hosted instance runs on a free tier that allows 512 MB of RAM. The pipeline does not fit, and the measurement is worth stating because it rules out the obvious workaround.

CONFIGURATION	PEAK RSS	READING
420 Delhi cells, full pipeline	1,083 MB	over budget by more than 2×
182 cells, reduced grid	701 MB	still over budget
Implied fixed cost	~283 MB	before a single cell is forecast
Python, pandas, XGBoost, boosters	~126 MB	the floor, with no data loaded
Snapshot mode, serving the bundle	130 MB	**comfortable, ~15 ms per request**

Table 10.1 Fitting a line through the two pipeline measurements gives a fixed cost of roughly 283 MB independent of grid size. Coarsening the domain would therefore have cost real resolution and still not fitted.

10.2 The bundle, and what it does and does not cost

The deployment serves a precomputed bundle of 20 routes captured from the same pipeline at full resolution: 1,120 cells, 2.8 km over Delhi, all four pollutants, the coupled solver, photolysis and the plume. The bundle is 805 KB and is captured from the running API rather than rebuilt from modules, so what ships is exactly what the endpoints return.

A deployment on this tier gives up freshness. It does not give up resolution, and it does not give up physics. That is the correct trade if one has to be made, and the dashboard states which mode it is in rather than presenting a stale forecast as current.

10.3 The trade no longer has to be made

The constraint is that the *web service* has 512 MB. Nothing requires the pipeline to run on the web service, and once that is noticed the problem dissolves.

1. A scheduled GitHub Actions workflow runs the full pipeline on a hosted runner. Standard runners provide four cores and **16 GB of RAM**, against a measured peak of 1.1 GB, so the pipeline runs at full resolution with an order of magnitude of headroom.
2. The workflow writes the bundle and **commits it back to the repository**.
3. Render **auto-deploys on push**, so the hosted site begins serving the fresh bundle within a couple of minutes.
4. Public repositories get **unlimited free Actions minutes**, so a six-hourly refresh costs nothing.

The workflow refuses to commit a bundle whose manifest contains a failed route, a suspiciously small payload, or a missing required one. Replacing a stale but correct forecast with a broken one is strictly worse than doing nothing.

Why a schedule is not an approximation of real time

No operational forecast system re-solves its physics when someone loads a page. Numerical weather prediction runs on a cycle and serves the most recent cycle's output, and a scheduled refresh is that same arrangement rather than a substitute for it. What remains is that a free instance sleeps after fifteen minutes idle, so the first request after a quiet period is slow. That is a cold start, not a stale forecast, and the two should not be confused.

A scheduled workflow on a repository with no commits is disabled after sixty days of inactivity. This one commits on every run, so it keeps itself alive.

11

What is not built, and why

Stated with the same care as the results. One clause remains partial, one headline result is negative, and several boundaries were chosen deliberately.

11.1 We do not run a chemical transport model C1 PARTIAL

There is no radiative transfer solve, no chemical mechanism integrated forward in time, and no gridded emissions inventory. WRF-Chem needs a Linux cluster, an MPI and netCDF stack, and a district level emission inventory we do not have. What we do instead is consume the output of coupled models that are already running operationally: CAMS for the chemistry prior and the aerosol optical depth the feedback solver runs on, and the MoES DSS as the transport reference.

The statement's "or similar open-source coupled frameworks" admits this reading, and there is genuinely a coupled framework in the pipeline. A reader who requires a CTM executed in house will not find one, and this report does not claim otherwise anywhere. **This is the one clause that stays partial by construction.**

11.2 The coupling does not improve forecast error, and that is reported as a result

The ablation has now been scored on the current models, and the honest answer is that the feedback does not earn its place on skill. Overall it makes the forecast **0.11 percent worse**. Section 8.5 gives the regime breakdown, which is the part worth reading: the sign flips exactly where the physics says it should.

The most likely explanation is the one prepared before the measurement was taken: the trained heads already capture most of this effect through their radiation, boundary layer and ventilation features, so an explicit solver adds little on top. That does not make the solver useless. It is what produces the counterfactual in §4.4, which no statistical model can answer, and it is what makes each mechanism separately auditable against published values. But it is not a skill driver, and presenting it as one would be false.

11.3 The DSS comparison is fairer, and still not matched on lead time

The caveat that our hindcast was driven by reanalysis has been removed rather than restated: §8.4 re-runs it on archived forecast runs, and the result did not degrade. Two smaller caveats remain and both favour us. The archive holds the best available run for each hour, which is a short lead time, while the DSS was scored 24 to 72 hours ahead. And boundary layer height alone is still backfilled from ERA5, because the forecast archive does not serve it.

There is also a confound worth keeping in view: the fair run gained a real vertical temperature profile at the same time it lost the reanalysis, so the two changes cannot be separated from this experiment alone. §8.4 states that rather than reading the improvement as a clean win.

11.4 Deliberate boundaries

- **No VOC chemistry.** Delhi's winter ozone is VOC limited as well as radiation limited, and we have no VOC measurements. TROPOMI formaldehyde and NO_2 could give a regime map, but that requires orbit level netCDF processing. Only the radiation limited half is modelled, and the report says so wherever ozone appears.
- **A single chemical layer.** Vertical structure is diagnosed, not resolved. The lid and mixing depth are tracked in time and height and modulate surface concentration, but the chemistry is solved in one layer. This is the largest remaining simplification in the system.

- **No data assimilation.** Observations enter as features and as training truth, not through an analysis cycle.
- **The plume magnitude is partly a DSS emulator.** One multiplicative scale factor is fitted against the DSS attribution. The physics parameters are not. So the correlation is an honest result and the magnitude is a calibration, and §6.2 reports them as two different things.
- **Three of four species respond without driving.** §4.5 explains why that is the physically correct arrangement, but it is a modelling choice and it is recorded as one.

12

What was closed, and what is left

An earlier draft of this report listed four outstanding items. All four have now been acted on. This section records what each cost and what it bought.

ITEM	STATUS	OUTCOME
All four species into the loop	DONE	Closes C6. PM10, NO_2 and O_3 now respond to the coupled meteorology, each by its own mechanism and each with its own acceptance bound. Eight of eight gates pass. §4.5
Scheduled snapshot refresh	DONE	Closes C7. The pipeline runs on a GitHub runner with 16 GB against a 1.1 GB peak, commits the bundle, and Render auto-deploys. Costs nothing. §10.3
Score the coupling ablation	DONE	Ran, and came out negative at −0.11 %. Reported as negative, with the regime split that makes it informative. §8.5 and §11.2
Fair comparison against the DSS	DONE	The blocker was assumed to be archive access. It was not: an archive of past forecast runs exists and is free. §8.4

Table 12.1 Of the eleven clauses in Table 1.1, ten are now met in full.

12.1 What the ablation result should change about how this is read

A negative result on the headline mechanism deserves more than a footnote. The coupling is physically correct, converges, and lands every magnitude inside published Delhi ranges it was never fitted to. It also does not measurably improve the forecast. Both of those are true at once, and the second does not cancel the first.

What the solver buys is not accuracy. It is the ability to answer questions a fitted model cannot: what happens to ozone if the aerosol halves, how much of tonight's PM2.5 is feedback rather than emission, and whether each link in the chain is individually defensible against the literature. Those are the questions the problem statement is actually about. A system that improved error by an unexplainable two percent would be worth less than one that explains why the air behaves as it does.

12.2 What is genuinely left

- **A true 72 hour lead comparison.** The archived forecast runs we can reach are short lead. Scoring against the DSS at matched lead needs an archive indexed by initialisation time, not by valid time.
- **Multi layer chemistry.** Resolving two or three layers rather than one would let the plume arrive aloft and mix down explicitly rather than through a gate.
- **TROPOMI derived VOC regime map.** Formaldehyde to NO_2 ratio, with the threshold near 3.1, would make the ozone chemistry regime aware.
- **Assimilation of station observations** into the initial state rather than only into the feature set.
- **Ablate the three new species separately.** The ablation reported here isolates the radiative feedback on PM2.5, which is what it was built to do. The PM10, NO_2 and O_3 responses have not been scored on their own.

A Appendix: where every number lives

No figure in this report was typed by hand. Each was read from the file named below, and each of those files is produced by a script in the repository.

CLAIM	FILE	PRODUCED BY
Twelve head skill, Table 7.2	`models/metrics_multiwinter.json`	`scripts/train.py`
Leave one station out, Table 8.1	`models/loso.json`	`scripts/loso.py`
Coupling magnitudes, Table 4.1	`data/snapshot/coupling.json`	`vayuchakra/feedback.py`
Photolysis and ozone, Table 4.2	`models/ozone_sensitivity.json`	`scripts/ozone_sensitivity.py`
Plume calibration, Table 6.1	`models/plume_calibration_octnov.json`	`scripts/plume_calibrate.py`
GRAP probabilities, Table 8.2	`models/uncertainty_pm25_24h.json`	`scripts/train_uncertainty.py`
DSS head to head, Table 8.3	`models/validation.json`	`scripts/validate.py`
Grid construction, Table 3.1	`data/snapshot/health.json`	`vayuchakra/grid.py`
Species responses, Table 4.3	`data/snapshot/coupling.json`	`vayuchakra/feedback.py`
Coupling ablation, Table 8.4	`models/validation.json`	`scripts/validate.py`
Abandoned UV check, §8.4	`vayuchakra/photolysis.py`	`cloud_attenuation_check()`

Table A.1 Every claim traced to its artefact.

A.1 Running it

- `pytest` runs 84 tests. They include a mesh walk asserting the two grid tiers leave no coverage gap, and they gate on every literature bound in Table 4.1, including a test that fails if a species is handed to the solver and never judged.
- `DECISIONS.md` records 61 decisions with the measurement behind each, negative results included.
- `scripts/run_all.sh` runs the pipeline end to end.
- The dashboard reads the same API this report cites, so anything on screen traces back the same way.

A.2 Sources cited

- Xing et al., *Impacts of aerosol-photolysis interaction and aerosol-radiation feedback on surface-layer ozone in North China*, Atmos. Chem. Phys. 22, 4101 (2022).
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- *The impact of aerosols on photolysis frequencies and ozone production in Beijing 2012 to 2015*, Atmos. Chem. Phys. 19, 9413 (2019).
- Master Chemical Mechanism v3.3.1 photolysis parameters, University of York.
- *Assessing CAMS reanalysis AOD over India*, Environ. Sci. Pollut. Res. (2025).
- MoES and IITM WRF-Chem Decision Support System (JAMES). Third party output, cited and never redistributed.
- CPCB National Air Quality Index, published breakpoint table.